

DENTINE

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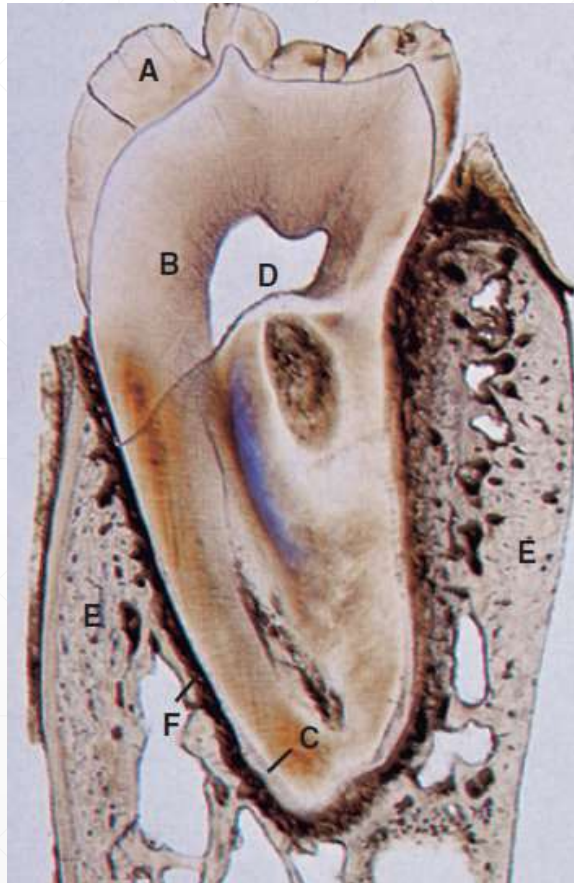
LECTURER- DEPT OF ORAL BIOLOGY



LEARNING OBJECTIVES

- By the end of this lecture, students will be able to;
 1. Describe the structure and composition of dentine, including its organic and inorganic components.
 2. Identify the histological features of dentine, including dentinal tubules and odontoblastic processes.
 3. Classify dentine according to time of formation and structure/location.
 4. Differentiate between primary, secondary, and tertiary dentine.
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WHAT IS DENTINE?



- Dentine is the mineralized tissue that forms the bulk of the tooth.
- Developmentally, it is derived from the dental lamina.
- It has a large number of small, parallel tubules in a mineralized collagen matrix. The tubules contain the long odontoblastic processes and extracellular (dentinal) fluid.
- Dentine is sensitive and is formed throughout life, giving rise to secondary dentine.
- The cell responsible for forming dentine is an odontoblast, and the cell body lies at the interface with the dental pulp: dentine–pulp complex.

PHYSICAL PROPERTIES

Dentine is pale yellow in colour and contributes to the appearance of the tooth through the translucent enamel.

Dentine is harder than bone and cementum, but softer than enamel.

Its organic matrix and tubular architecture provide it with greater compressive, tensile and flexural strength than enamel.

Dentine is permeable, the permeability depending on the size and patency of the tubules, which will decline with age.

CHEMICAL PROPERTIES

Dentine consists of apatite crystals on an organic scaffold predominantly composed of collagen.

The inorganic mineral component is in the form of calcium hydroxyapatite crystals. The crystallites are calcium-poor and carbonate-rich.

The hydroxyapatite crystallites in the mineralised dentine are found on and between the collagen fibrils.

COMPOSITION BY WEIGHT

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graph TD; A[COMPOSITION BY WEIGHT] --- B[70% - inorganic material<br/>(hydroxyapatite crystals)]; A --- C[20% - organic material<br/>•90% collagen (type 1)<br/>•8% non collagenous proteins, 2% lipids]; A --- D[10% - water]
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70% - inorganic material
(hydroxyapatite crystals)

20% - organic material

- 90% collagen (type 1)
- 8% non collagenous proteins, 2% lipids

10% - water

The **inorganic mineral component** is in the form of calcium hydroxyapatite crystals. The crystals are found between the collagen fibrils and are less densely packed. They are calcium-poor and carbonate-rich.

The **organic matrix** consists of collagen fibrils (type I collagen) embedded in an amorphous ground substance. The fibers run parallel to the pulpal surface and are of a larger diameter (100 nm).

Non-Collagenous Proteins

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graph TD; A[Non-Collagenous Proteins] --> B[Dentine Phosphoproteins]; A --> C[Dentine Matrix Proteins (DMP-1)]; B --> D["Dentine Phosphoprotein (DPP)"]; B --> E["Dentine Matrix Proteins (DMP-1)"]; D --> F["-Highly phosphorylated: acidic"]; D --> G["•-Binds Ca²⁺"]; D --> H["•-Controls crystal growth"]; E --> I["-Initiates mineralization"]; E --> J["-Cell attachment"]; E --> K["-Abundant in peritubular dentine"];
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Dentine Phosphoproteins

Dentine Phosphoprotein (DPP)

- Highly phosphorylated: acidic
 - Binds Ca^{2+}
 - Controls crystal growth

Dentine Matrix Proteins (DMP-1)

- Initiates mineralization
- Cell attachment
- Abundant in peritubular dentine

Other Non-Collagenous Proteins

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graph TD; A[Other Non-Collagenous Proteins] --> B[Proteoglycans<br/>(biglycan and decorin)]; A --> C[Glycoproteins]; A --> D[Gla Proteins]; A --> E[Growth Factors<br/>(TGF-β, BMP, IGF)];
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Proteoglycans (biglycan and decorin)

- Organizes collagen
- Cell adhesion

Glycoproteins

- Osteonectin
- Osteopontin
- Dentine sialoproteins

Gla Proteins

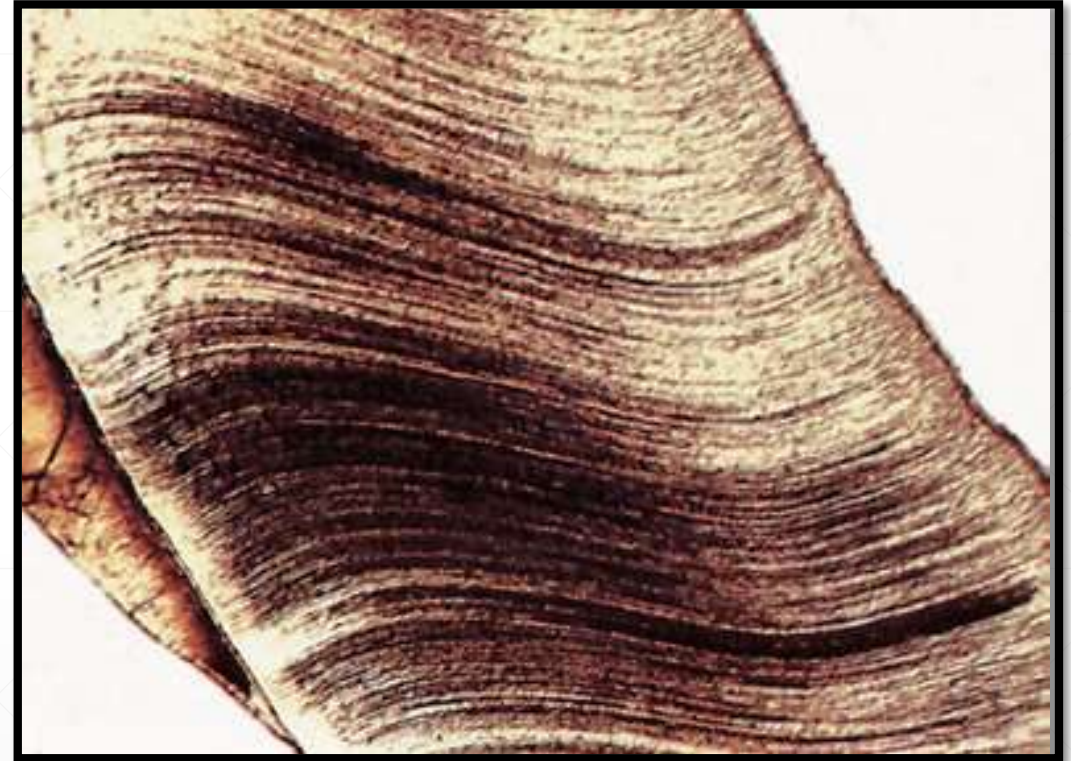
- Bind hydroxyapatite

Growth Factors (TGF- β , BMP, IGF)

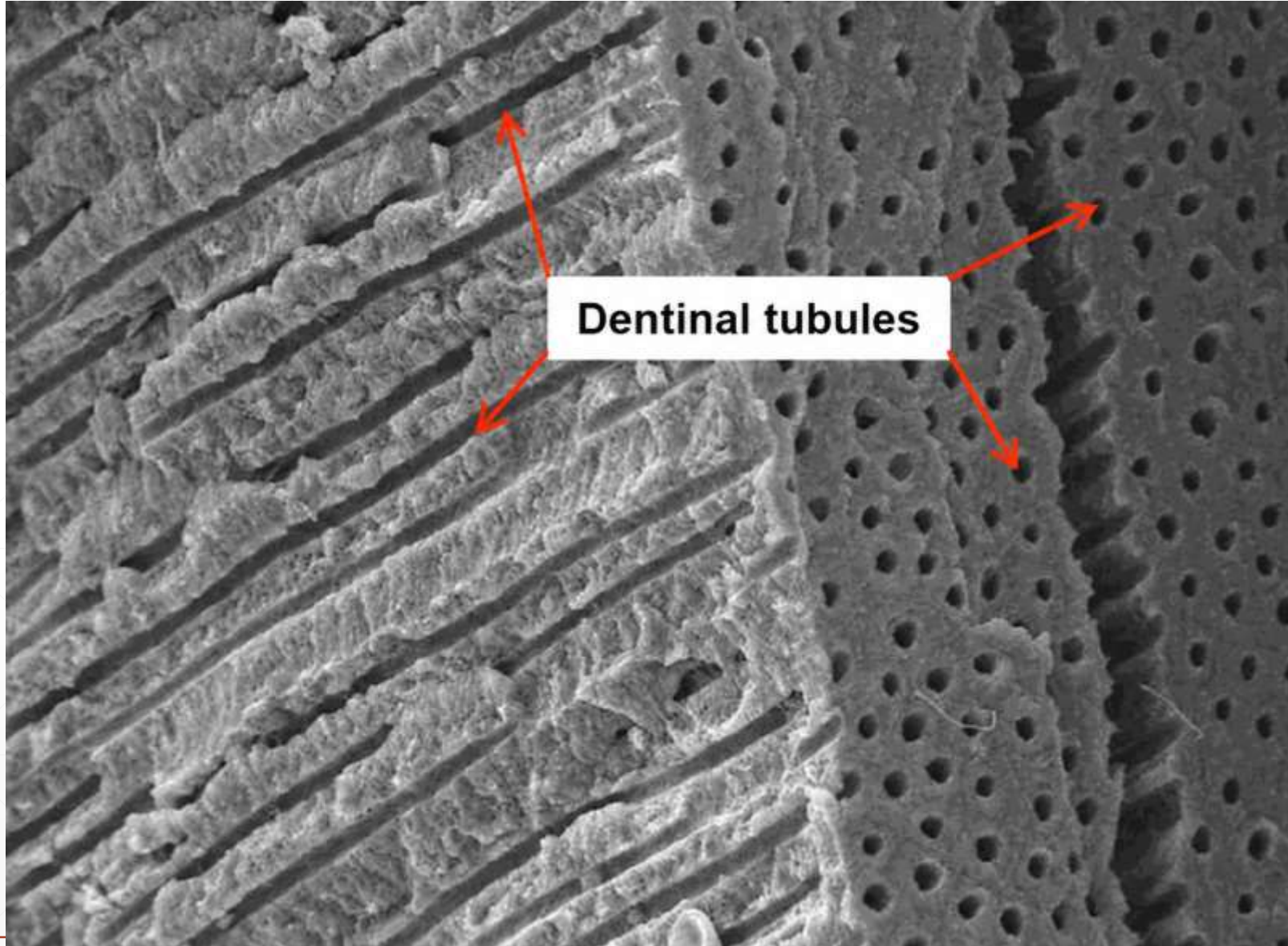
- Repair and regeneration

DENTINAL TUBULES

They run from the pulpal surface to the enamel–dentine and cementum–dentine junctions. Following a curved, sigmoid course: the primary curvatures.

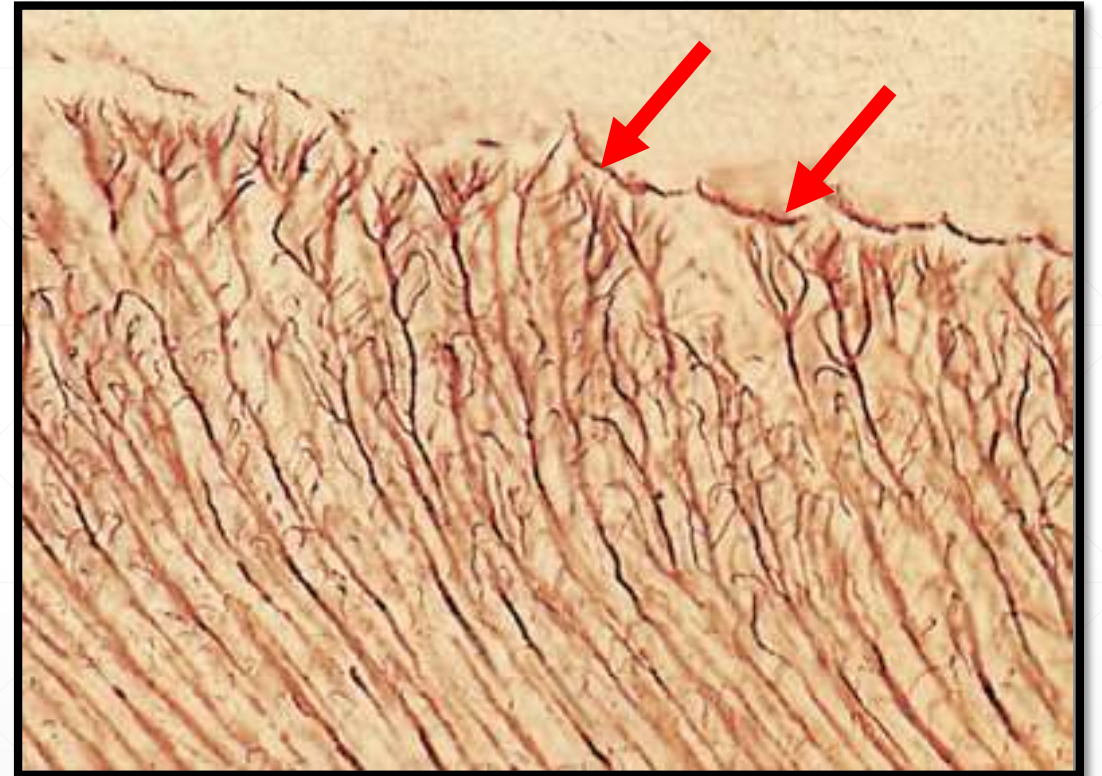


Ground section of dentinal tubules cut longitudinally, showing sinusoidal primary curvatures



DENTINAL TUBULES

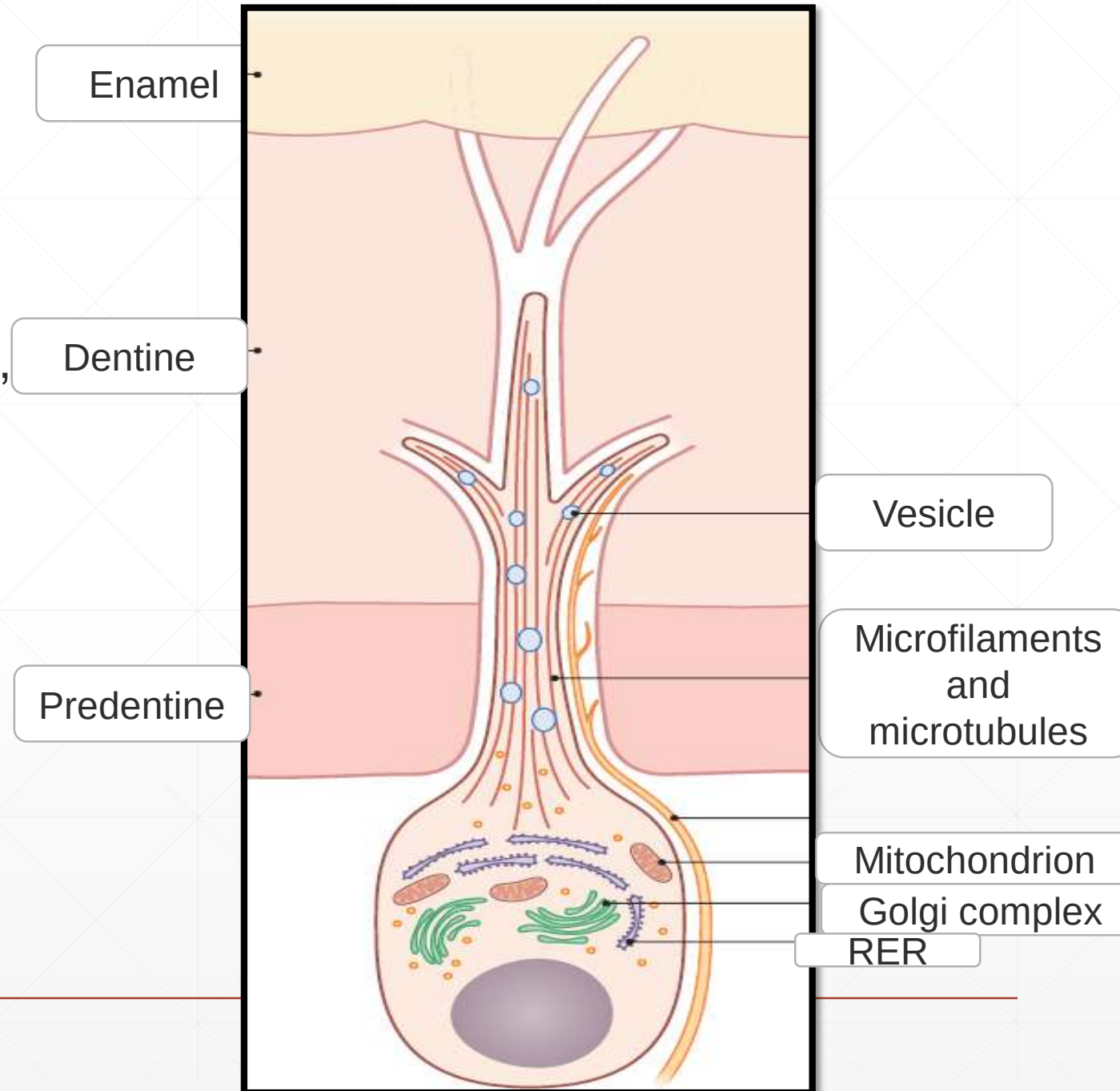
Dentinal tubules exhibit branching, and they branch profusely in the periphery near DEJ. In the root, the terminal part of the tubule branches, and the branches loop. This looping is responsible for the appearance of the **Granular Layer of Tomes**.



Ground section of dentine near DEJ showing branching of dentinal tubules

DENTINAL TUBULES

- An odontoblast cell and its odontoblastic process (Tomes' fiber) extending through a dentinal tubule from the pulp toward the enamel.
- The cytoplasmic extension of the odontoblast extends through the dentinal tubule and is surrounded by dentinal fluid.

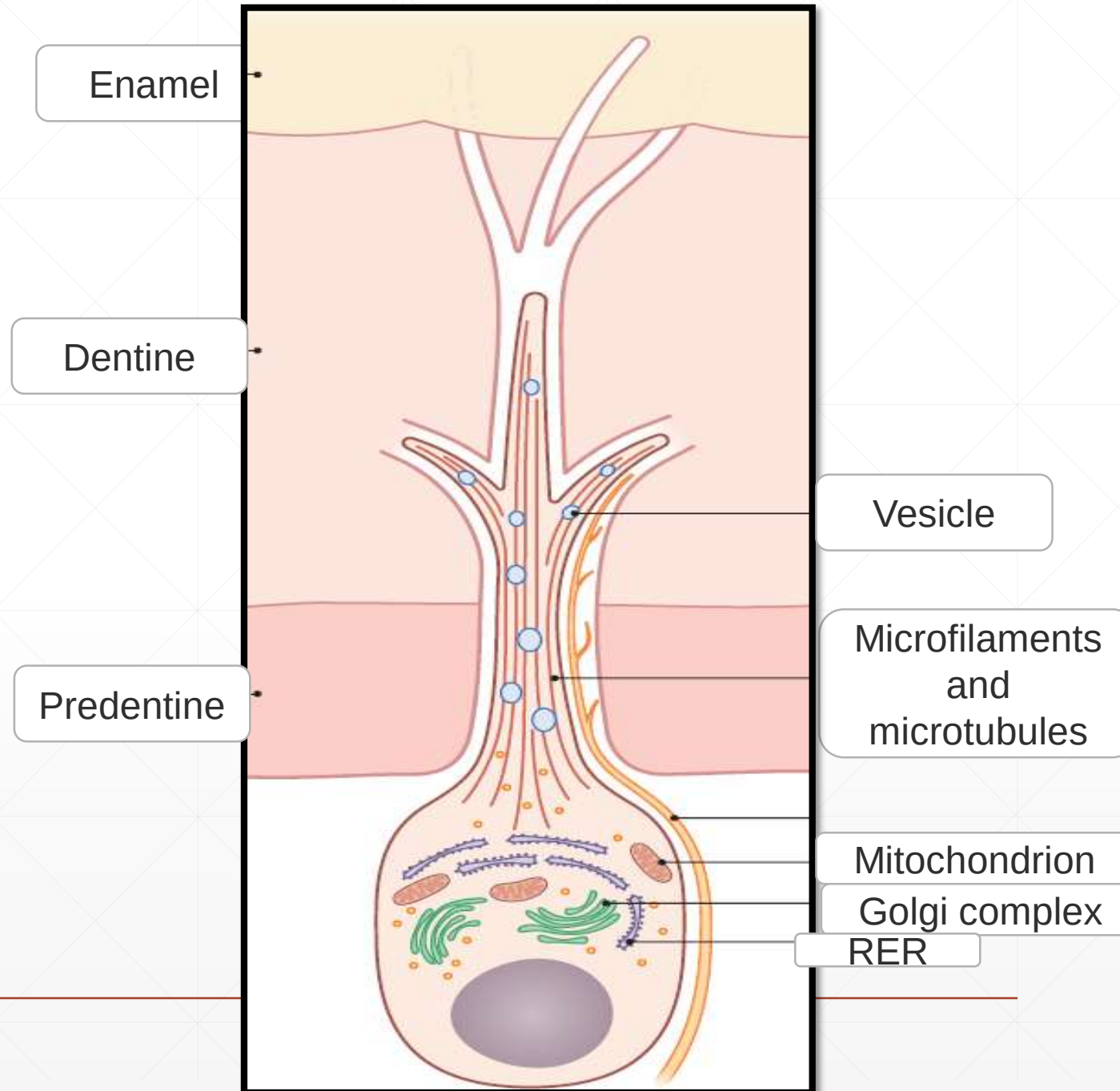


DENTINAL TUBULES

- Contents of dentinal tubules: odontoblastic process, dentinal fluid, nerve fiber.
- Microtubules/filaments within the odontoblastic process:

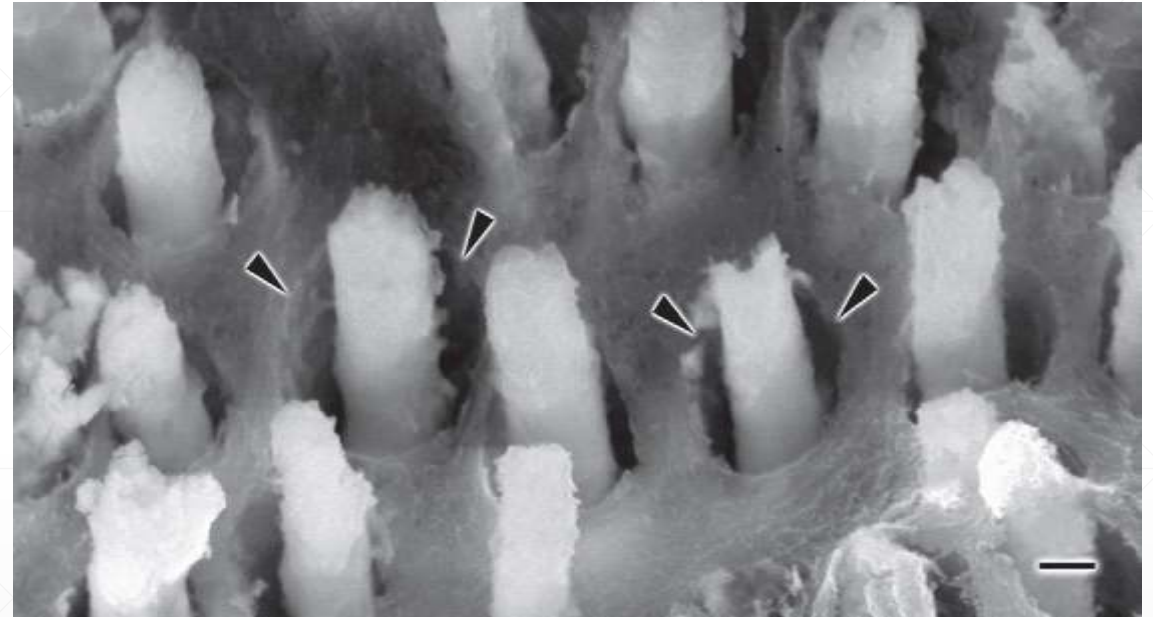
-**Tubulin**: microtubules helps in transport of vesicles

-**Actin**: microfilament helps in contraction hence Helps secretion of dentine matrix



DENTINAL TUBULES

- A proteinaceous membrane termed the **lamina limitans** lining the wall of the dentinal tubule. It primarily has proteoglycans that is composed mainly of glycosaminoglycans.

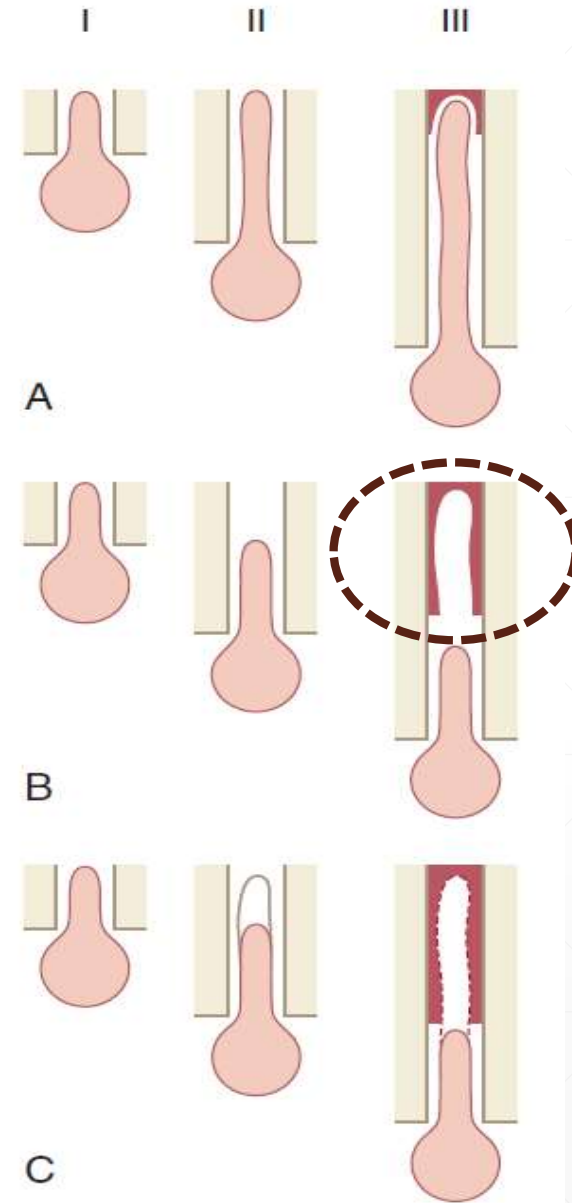


FATE OF ODONTOBLAST PROCESS

A- the process grows in length as dentine is deposited, and its peripheral termination remains at the outer end of the tubule.

B- the process reaches a predetermined length and then moves pulpally as dentine is formed, leaving behind an empty tubule.

C- the peripheral end of the processes degrades sequentially, and its remains form part of the matrix for the peritubular dentine.



Postodontoblastic space is from which the process has receded

CLASSIFICATION AND TYPES



According to
the time of
formation

According
regional
variations

According to
the pattern of
mineralization

ACCORDING TO TIME OF FORMATION

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graph TD; A[ACCORDING TO TIME OF FORMATION] --> B[Primary Dentine]; A --> C[Secondary Dentine]; A --> D[Tertiary Dentine];
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Primary Dentine

- Formed before eruption
- Forms the main bulk

Secondary Dentine

- Forms after eruption (root completion)
- It is formed throughout life

Tertiary Dentine

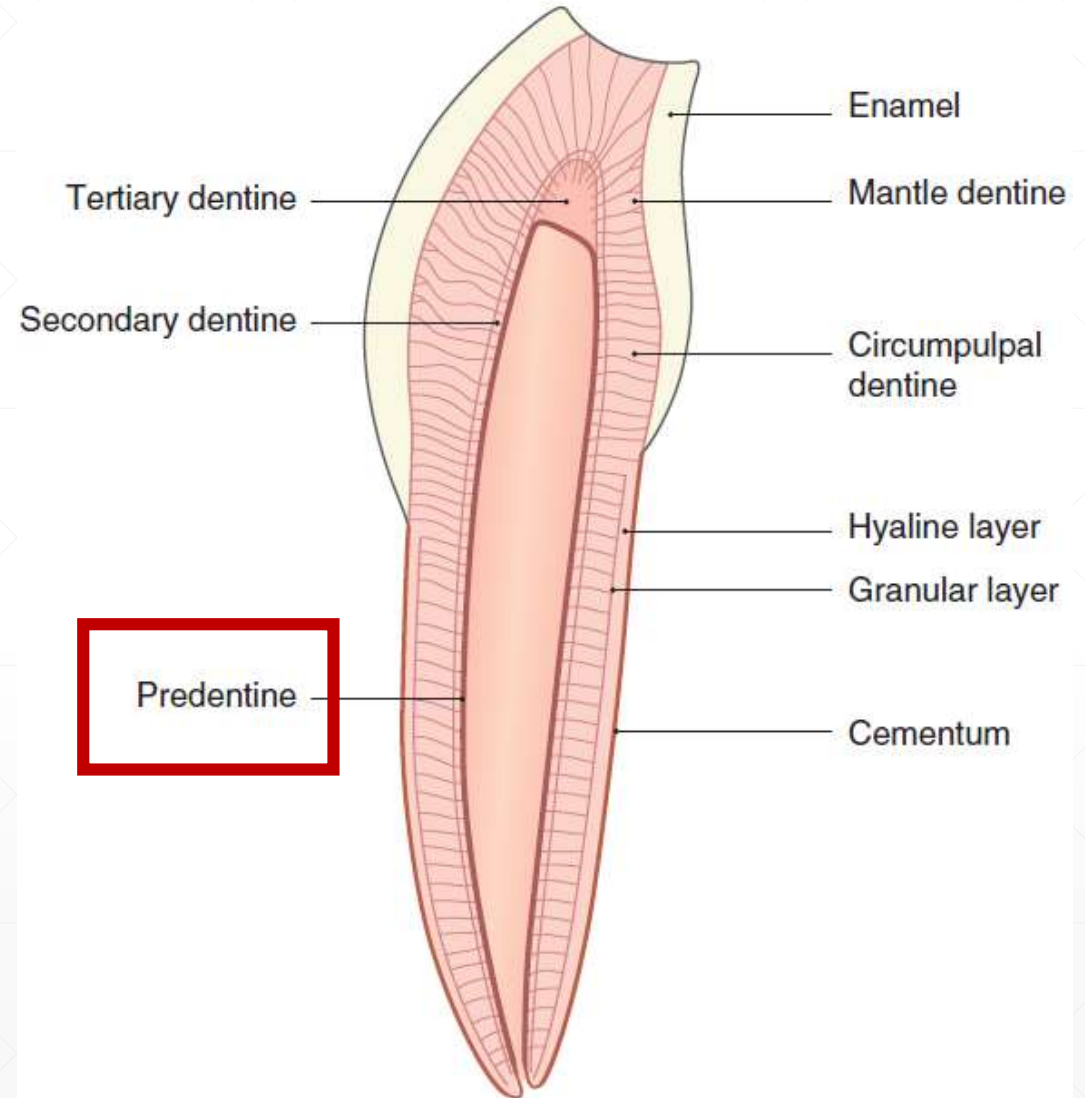
- Formed due to injury/insult
- 1. Reactionary
- 2. Reparative

ACCORDING TO REGIONAL VARIATIONS/HISTOLOGY

Type	Features
Mantle Dentine	First formed dentine beneath enamel
Circumpulpal Dentine	Main bulk surrounding pulp
Predentine	Unmineralized dentine adjacent to pulp
Peritubular Dentine	Highly mineralized around dentinal tubules
Intertubular Dentine	Between tubules; rich in collagen
Sclerotic Dentine	Tubules calcified/obliterated
Interglobular Dentine	Incomplete fusion of calcospherites

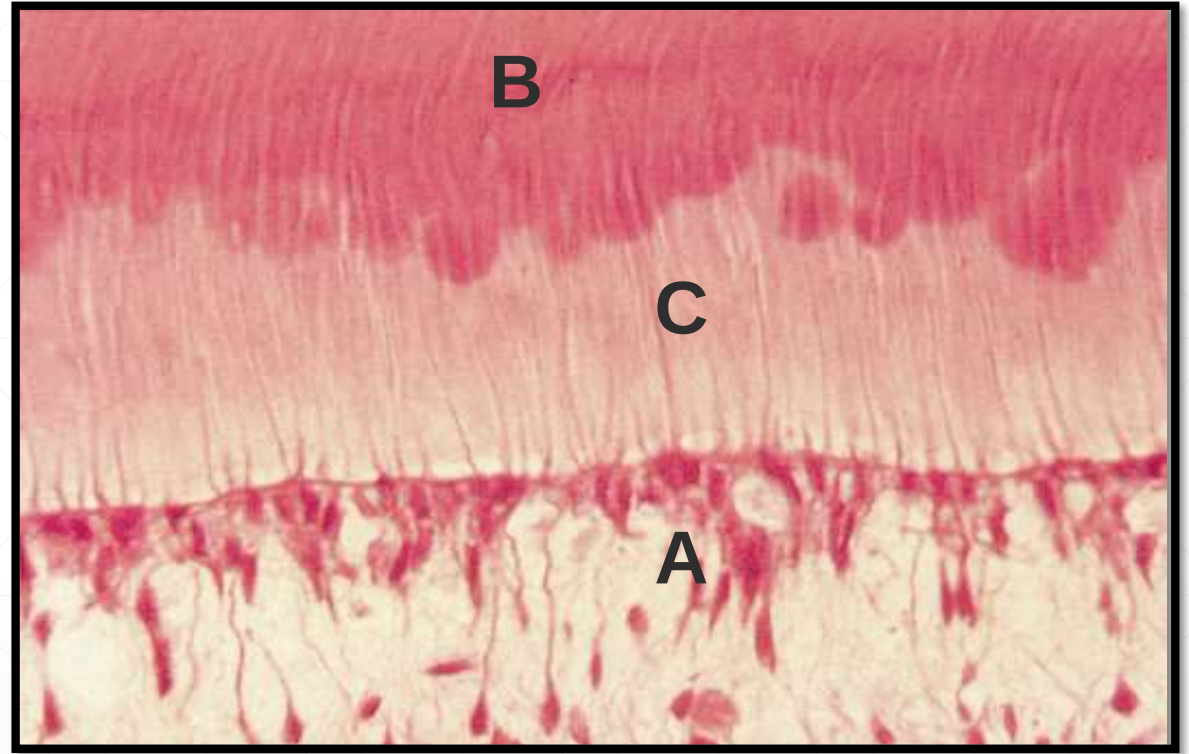
PREDENTINE

- Predentine is the innermost unmineralised layer (closest to pulp), where new dentine is being deposited throughout life.
- It is the initially laid down dentine matrix (organic) before its mineralisation.
- The width of the predentine can vary from 10 to 40 μm .



PREDENTINE

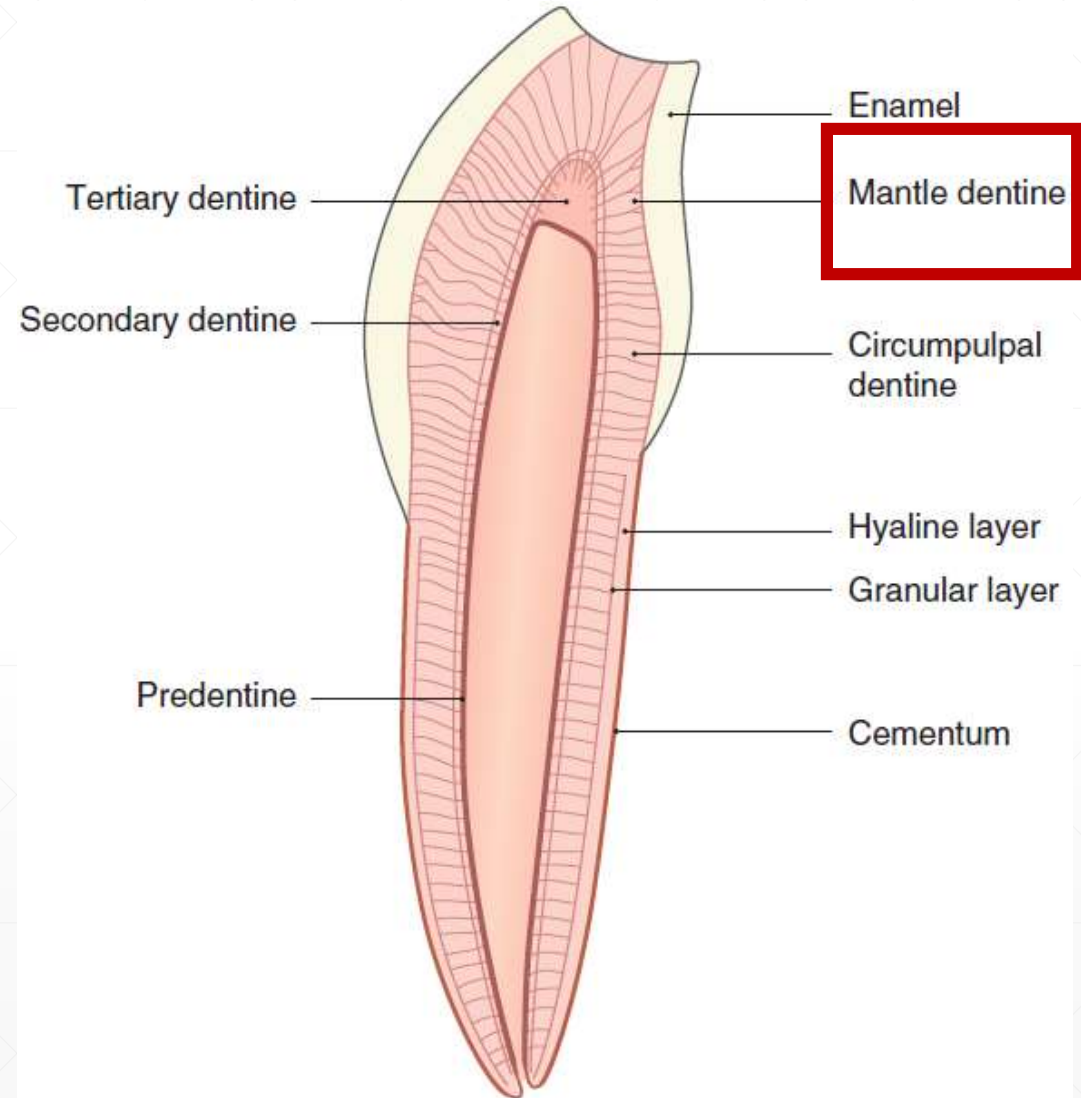
- Just peripheral to the predentine is a zone of mineralization.
- In the predentine, odontoblasts secrete the organic matrix. In mineralizing dentine, it participates in matrix modification.



Decalcified section of pulpodentinal region. A- Odontoblast layer. B- Mineralised dentine matrix. C- Predentine

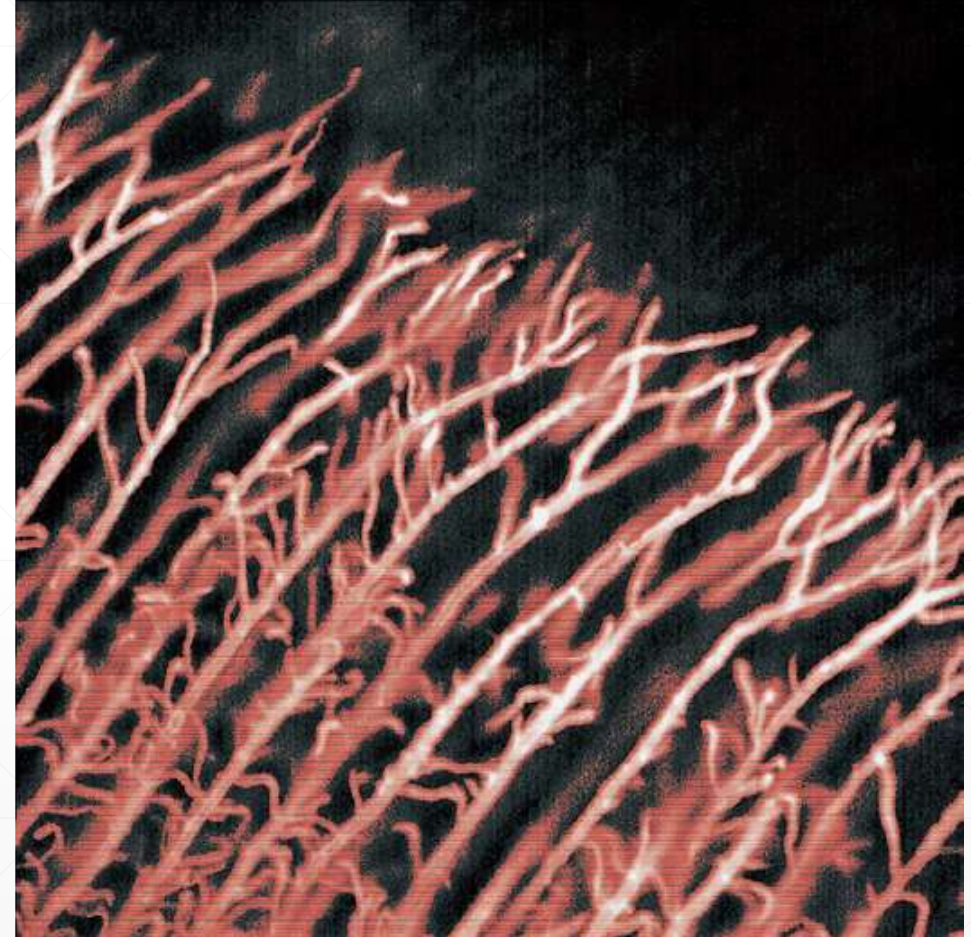
MANTLE DENTINE

- In the crown, this first-formed layer of dentine is known as mantle dentine (primary).
- It is 5% less mineralized than circumpulpal dentine.
- The mantle layer varies in width from 20 μm to 150 μm .



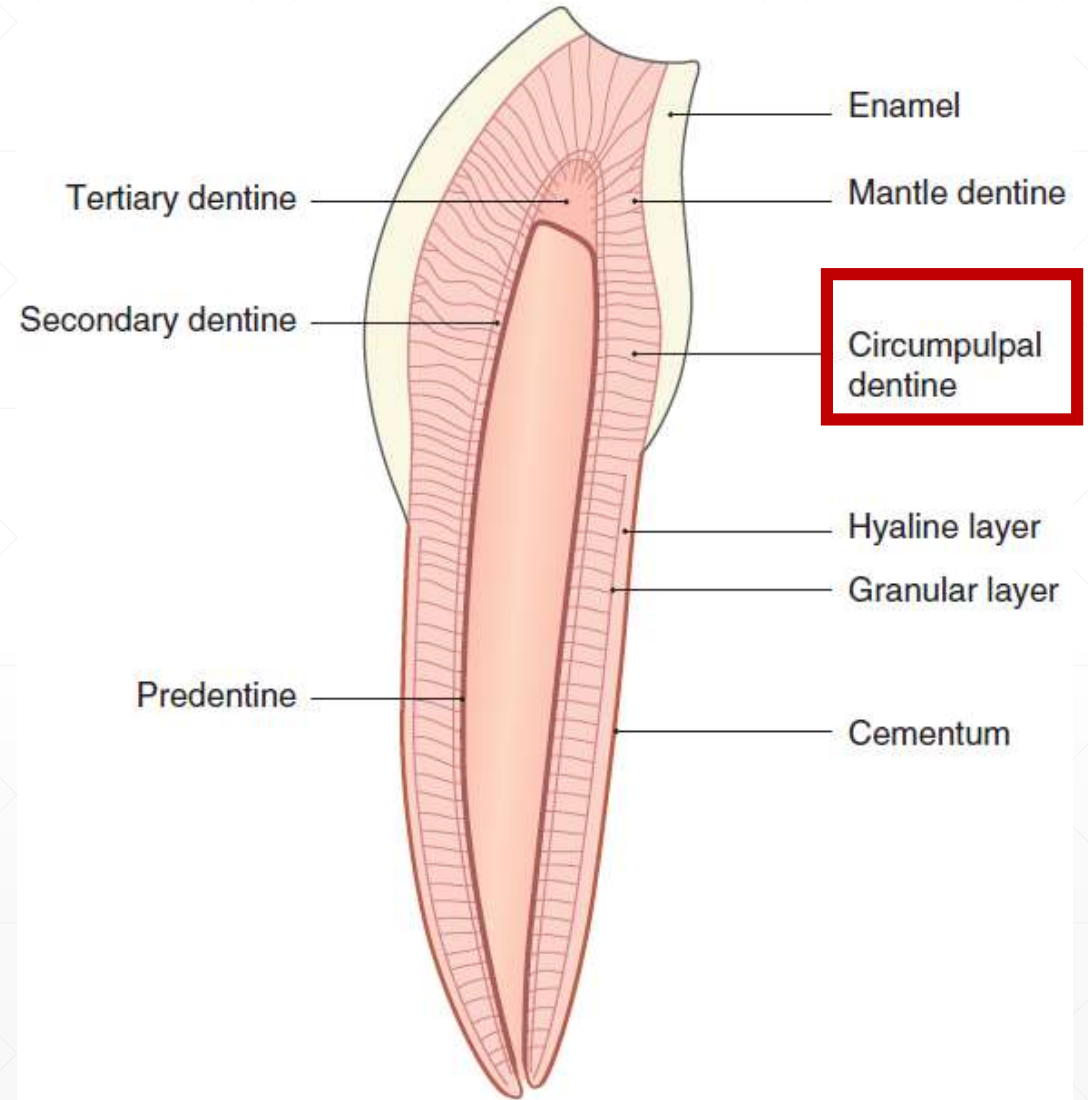
MANTLE DENTINE

- The collagen fibrils are oriented perpendicular to the enamel–dentine junction.
- The dentinal tubules branch profusely in this region.



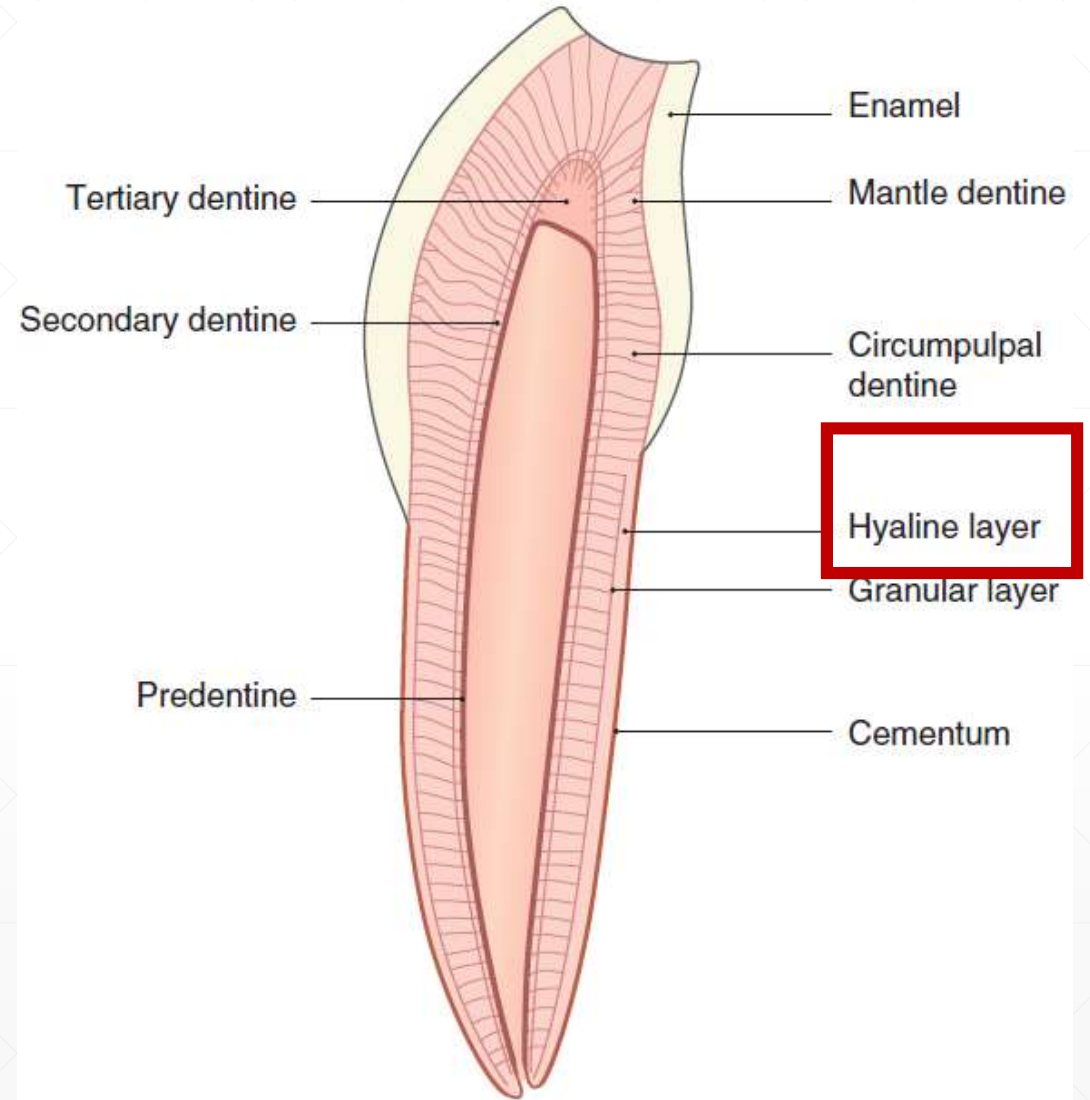
CIRCUMPULPAL DENTINE

- The bulk of the dentine in the crown and root between mantle and the zone of mineralisation is the circumpulpal dentine (also primary).
- The outer part is incompletely mineralised and has a characteristic appearance when seen in ground sections, referred to as 'interglobular dentine'.



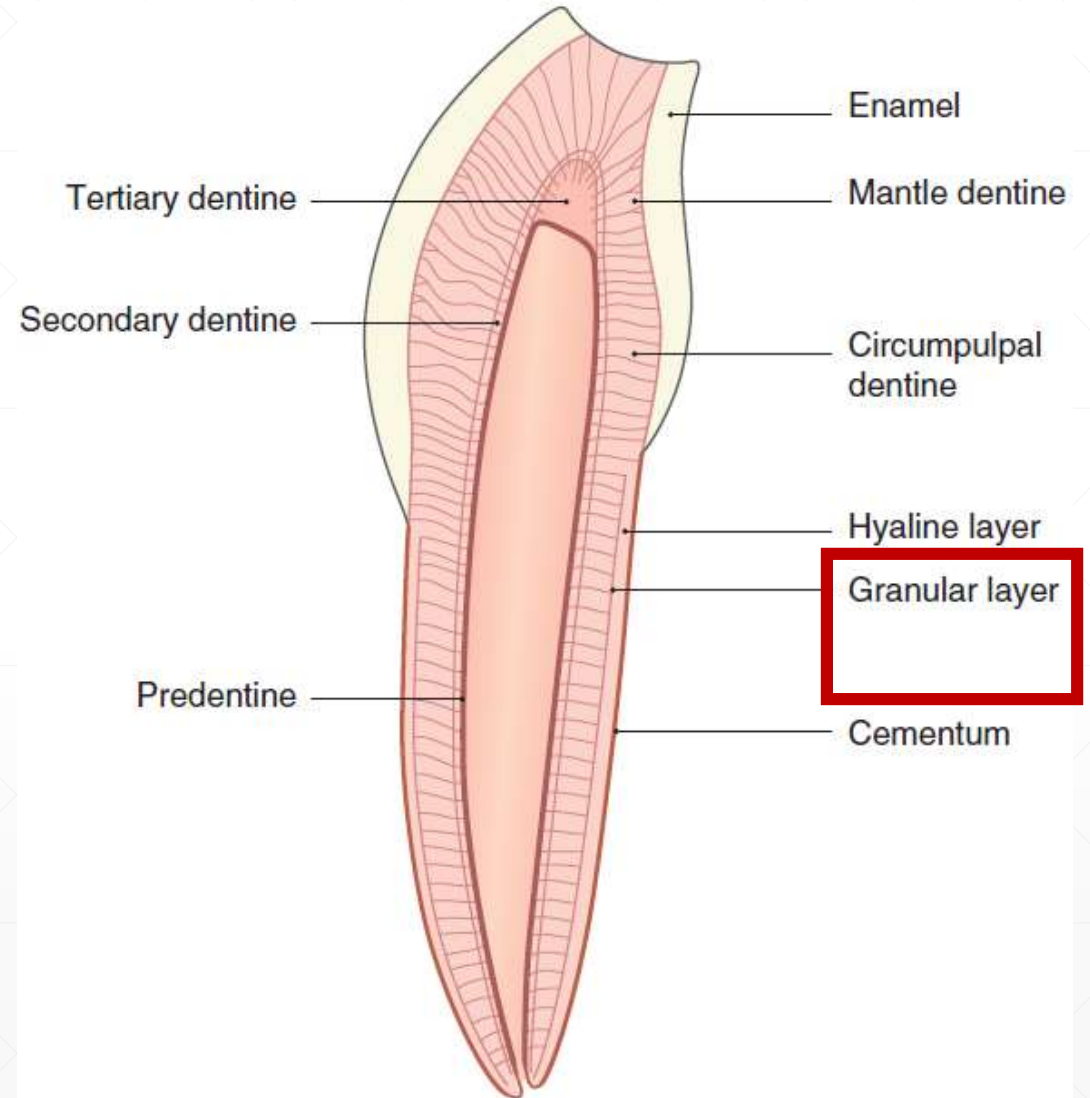
HYALINE LAYER

- The hyaline layer of Hopewell-Smith is a thin homogeneous layer present at the CDJ b/w root dentine and cementum.
- It is structureless, appears glassy, and lacks collagen fibres.



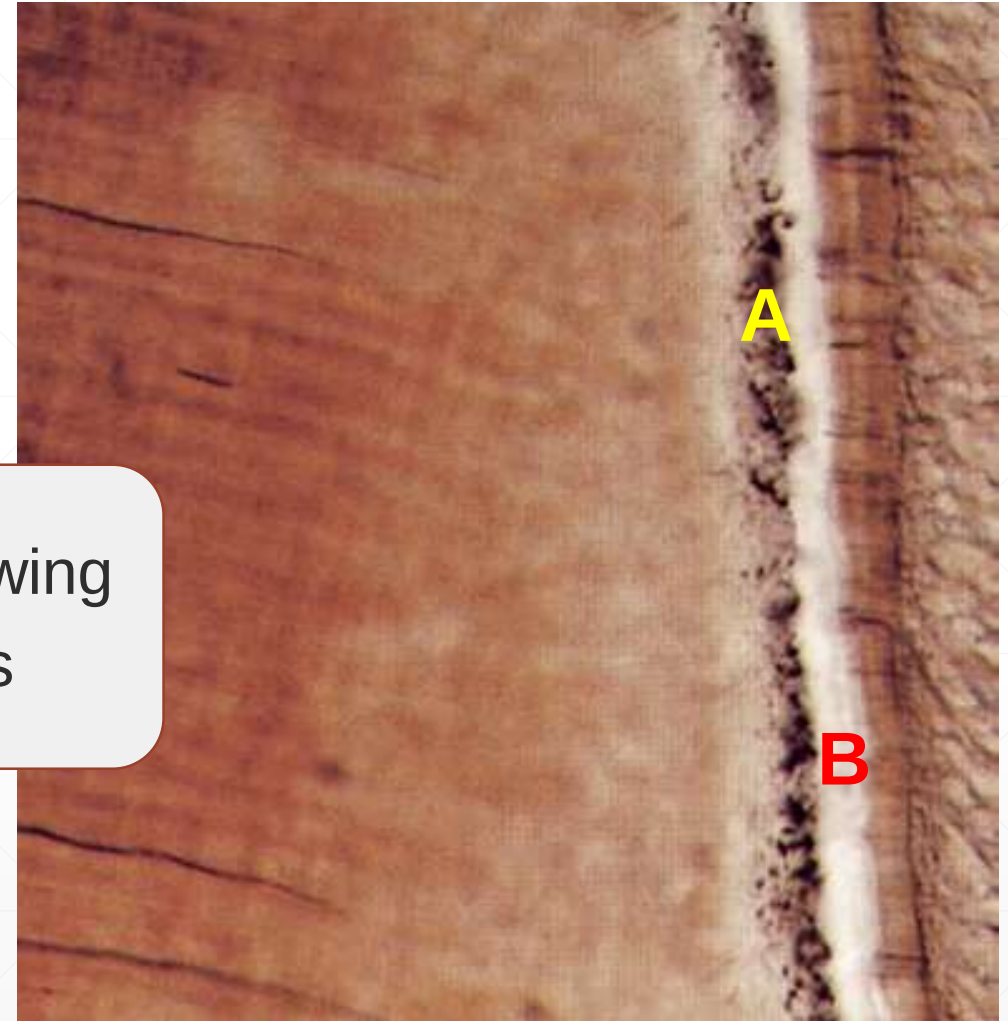
GRANULAR LAYER OF TOMES

- The periphery of dentine in the root has dentinal tubules branching more profusely and looping back on themselves, creating air spaces. These spaces appear empty, granular and hence dark in ground sections.



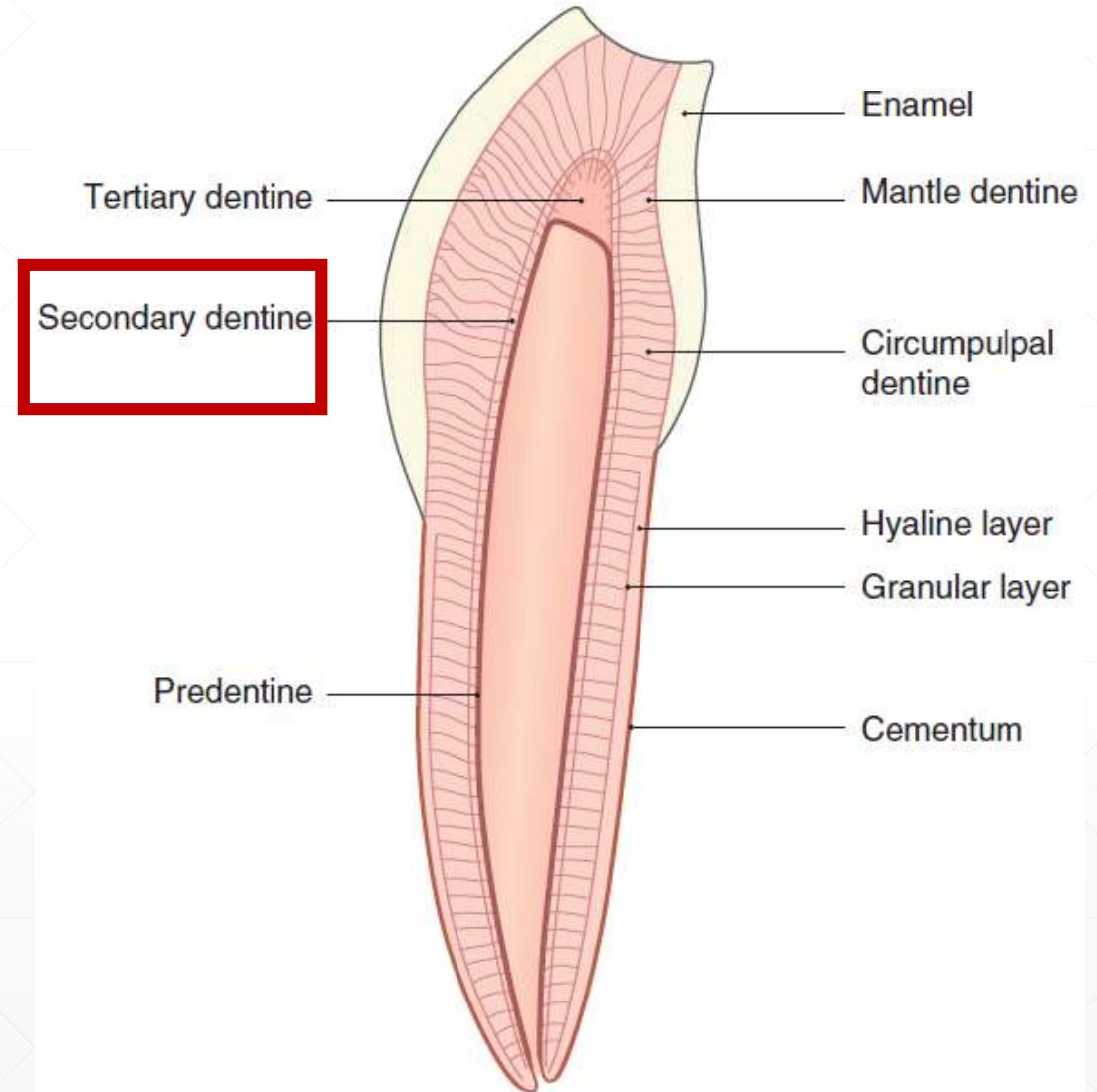
GRANULAR LAYER OF TOMES

Ground longitudinal section of a root showing the granular (A) and hyaline (B) layers



SECONDARY DENTINE

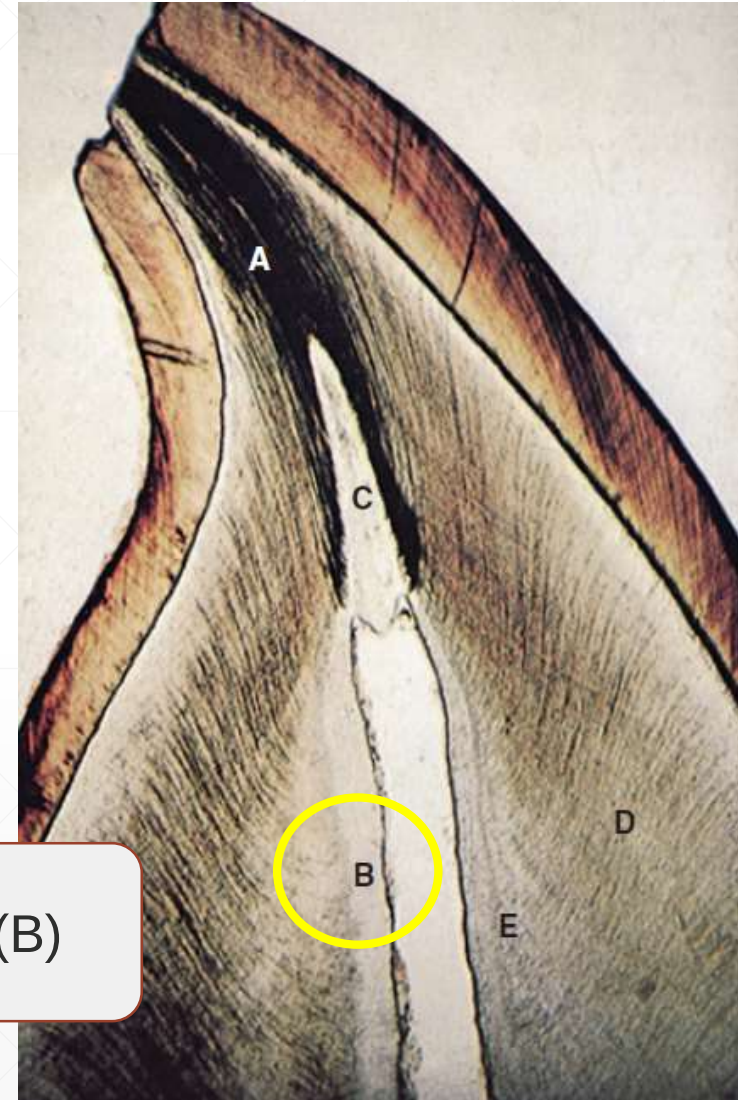
- Secondary dentine is the dentine formed after completion of root formation and closure of the apical foramen. It is formed slowly throughout life and produced by existing odontoblasts.



SECONDARY DENTINE

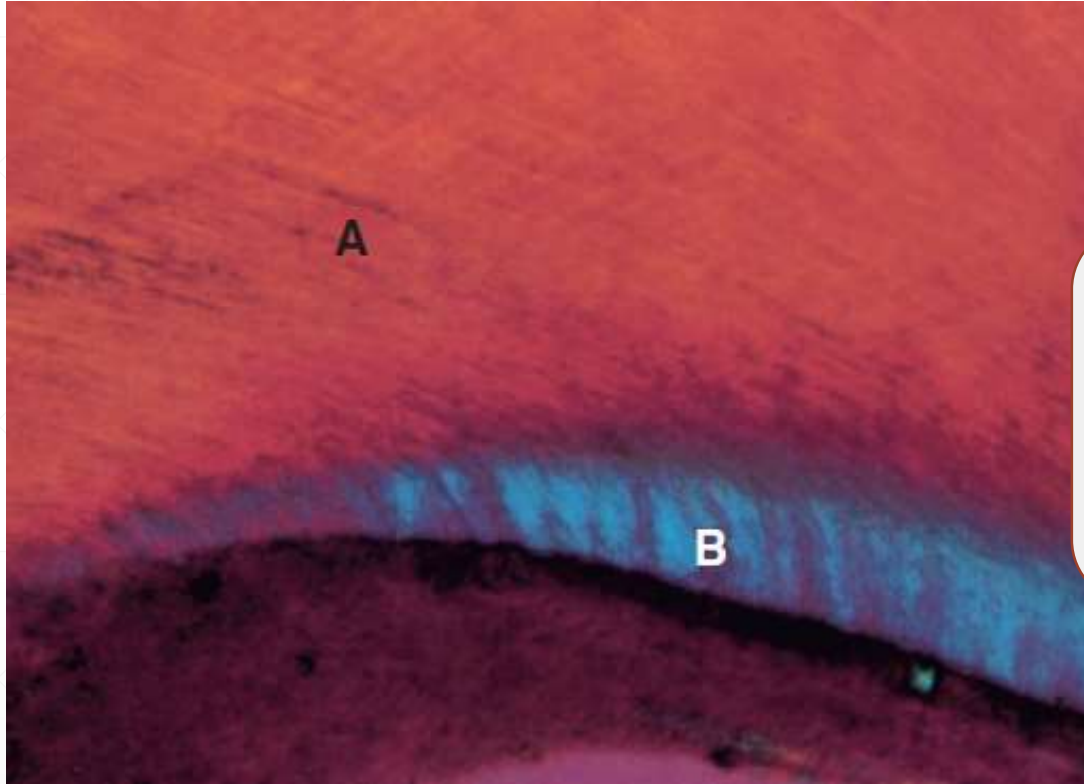
- It is most rapidly formed at the pulpal floor. Its continuing deposition leads to smaller pulp chambers and narrower root canals in older patients.

Ground section showing secondary dentine at (B)



Primary Dentine is formed during active tooth development. Collagen fibers are well-organized and regularly arranged. It produces a uniform tubular structure.

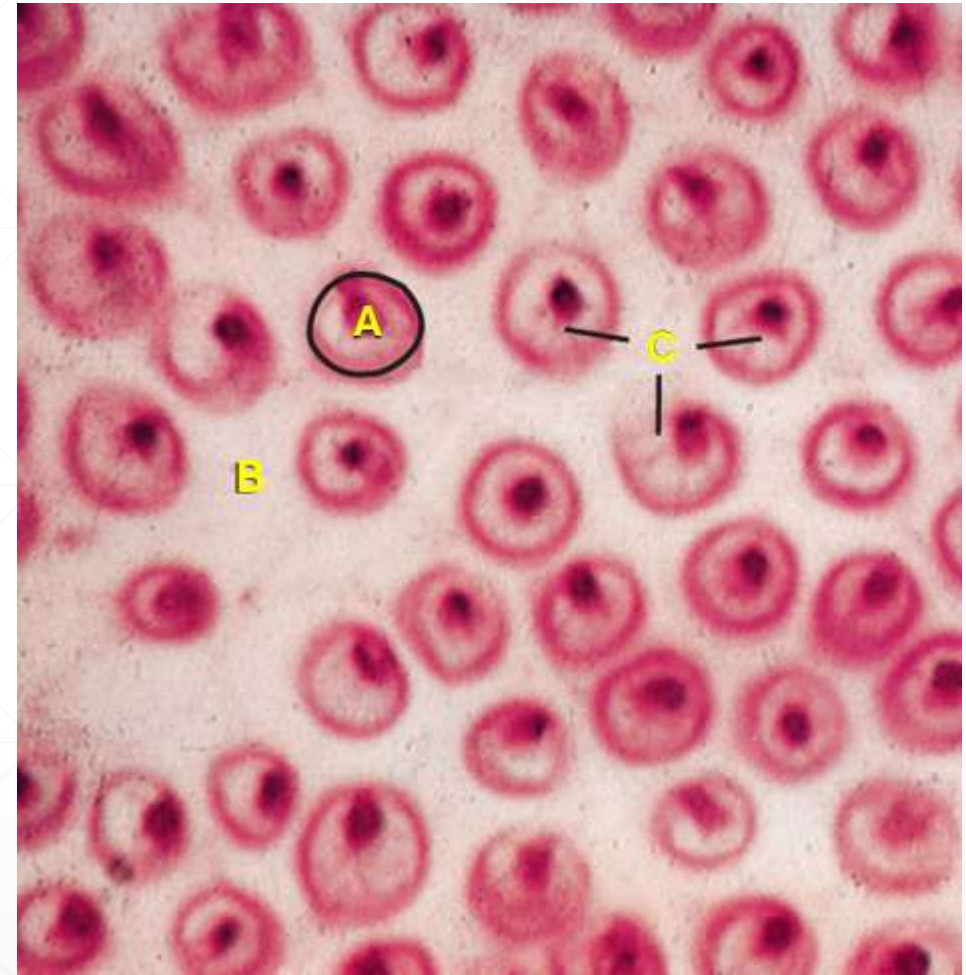
Secondary Dentine is slowly formed during physiologic aging. Collagen fibers become less organized. It produces an irregular tubular pattern.



The difference in colour between primary (A) and secondary (B) dentine is due to the difference in orientation of collagen

PERITUBULAR DENTINE

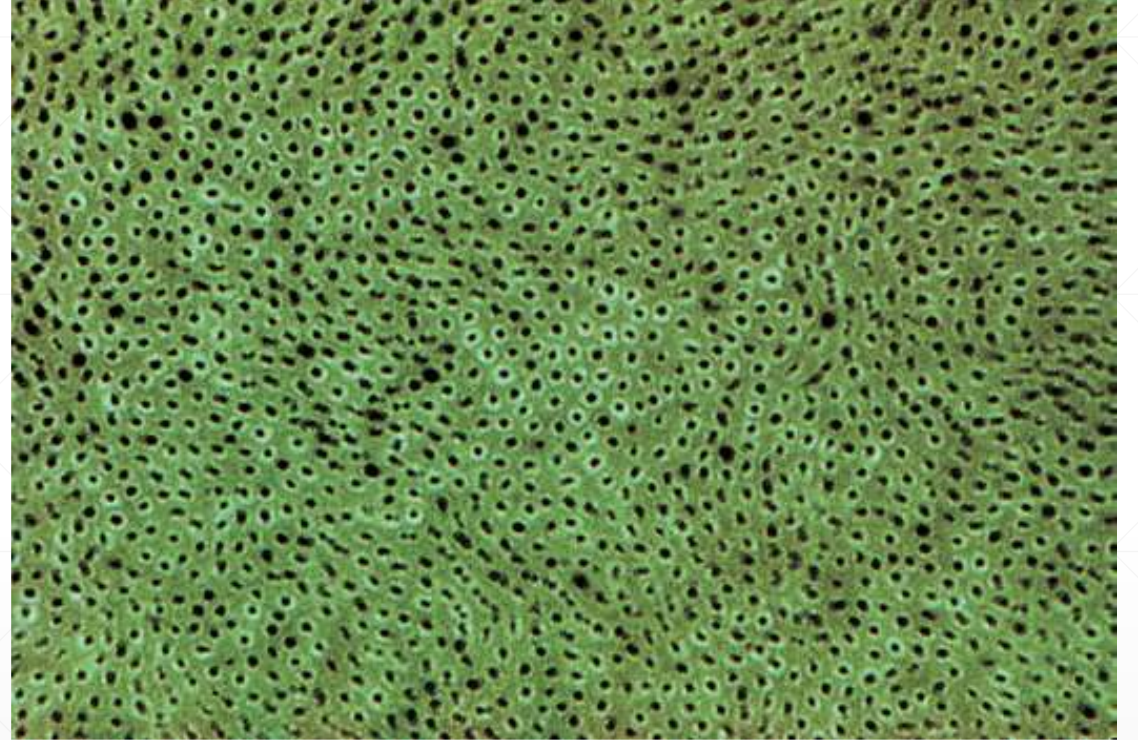
- **Intertubular dentine** is found between the dentinal tubules.
- The dentine deposited around the walls of the dentinal tubule is known as **peritubular dentine** (also known as intratubular dentine).
- Peritubular dentine is about 5–12% more mineralized than intertubular.



Demineralized section of dentinal tubules cut transversely. A-perimeter of a tubule, B-intertubular dentine, C-peritubular dentine.

PERITUBULAR DENTINE

- It is highly mineralized dentine, formed along the walls of dentinal tubules.
- It narrows the lumen, and its formation gradually leads to obliteration of the tubule.



Transversely sectioned dentinal tubules surrounded by radio-opaque and denser peritubular dentine

PERITUBULAR DENTINE

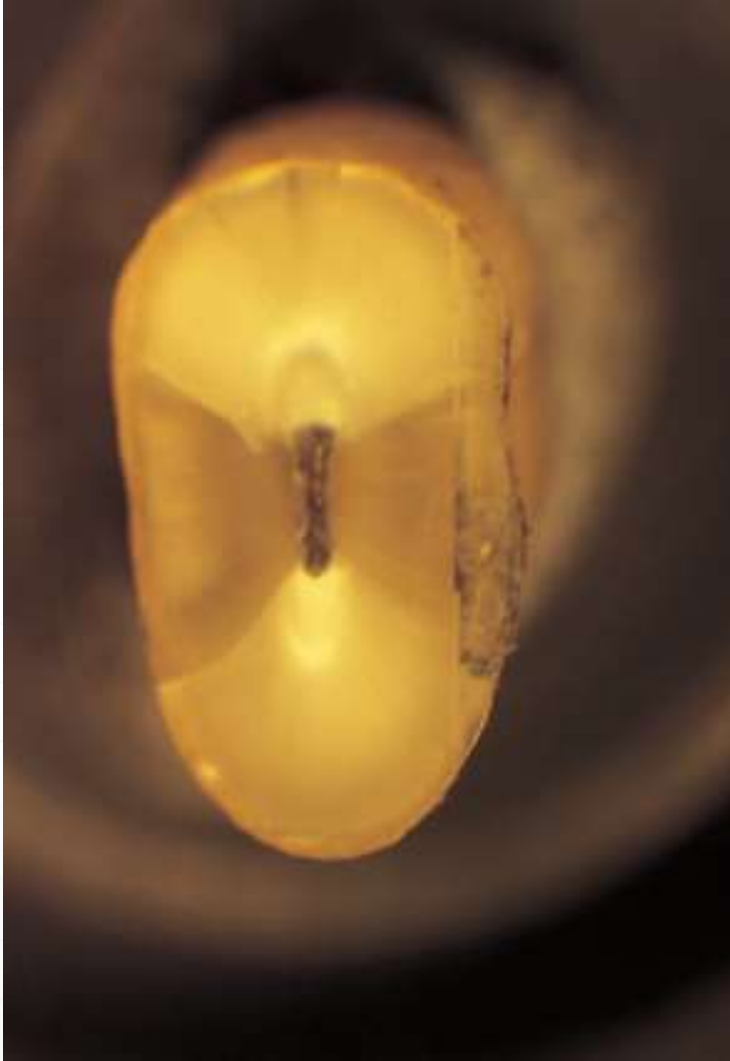
- When primary dentine formation is complete, all peripheral tubules have a lining of peritubular dentine that extends from the enamel–dentine junction.
- With physiological aging, dentinal tubules become completely occluded by peritubular dentine. When a ground section of a root is placed in water, regions blocked by peritubular dentine will appear translucent, ‘**translucent dentine**’.



Transversely sectioned dentinal tubules surrounded by peritubular dentine



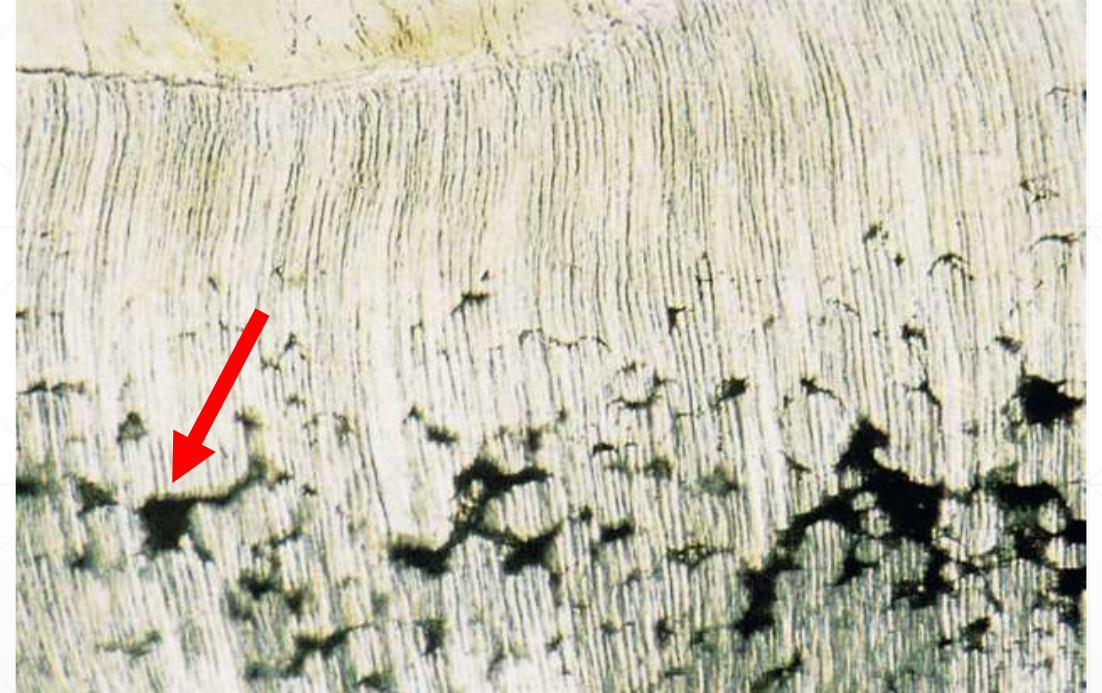
A section of a whole tooth showing the translucent dentine predominantly in the apical part of the root



In cross-section, translucent zones have a **butterfly shape** owing to the convergence of the tubules pulpally, being wider at the mesial and distal margins

INTERGLOBULAR DENTINE

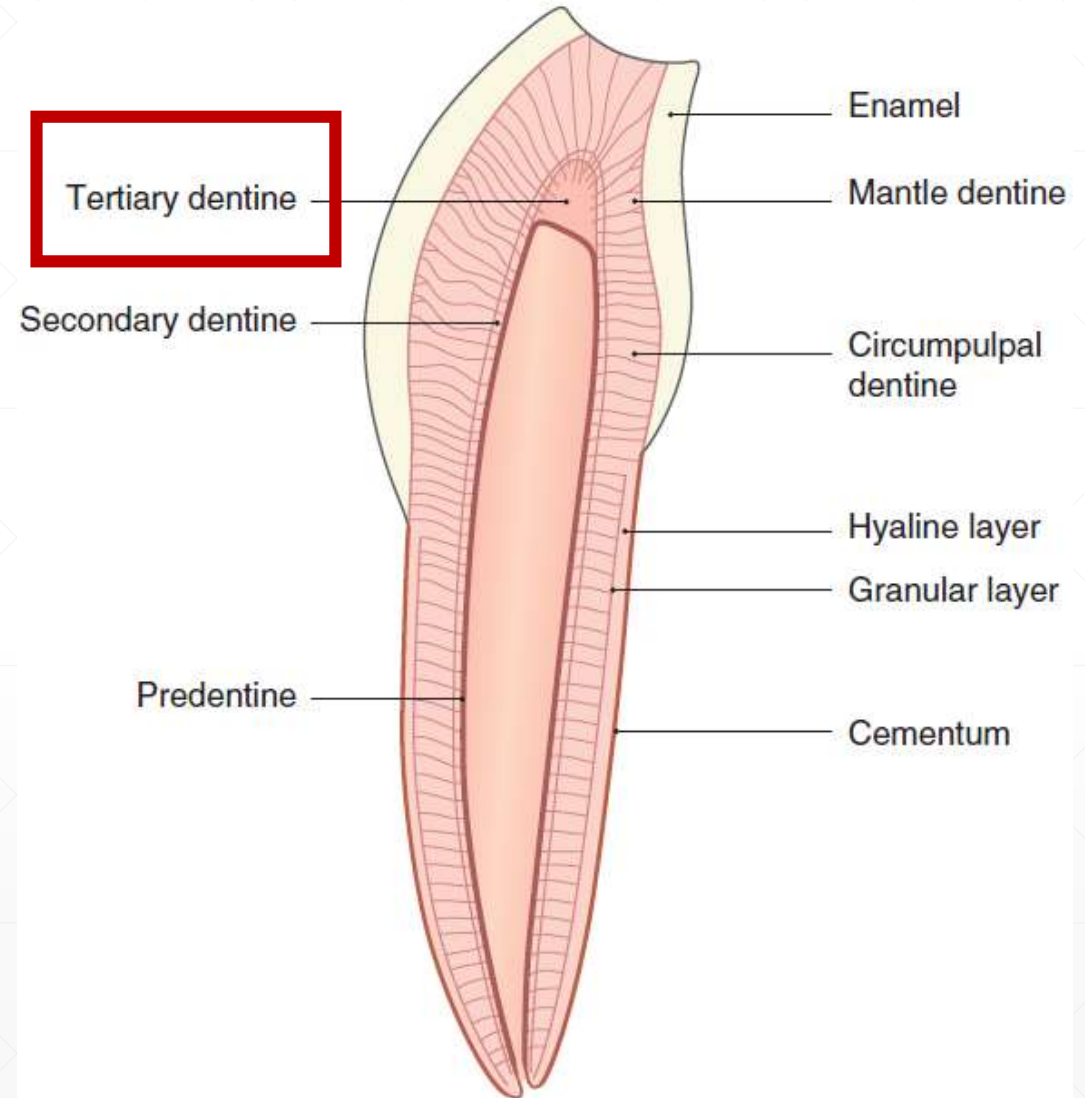
- Calcospherites are spherical zones of mineralization formed by hydroxyapatite crystals. Their fusion produces mineralized dentine, while failure to fuse results in **interglobular dentine**.
- It is present beneath the mantle layer in crown and beneath the granular layer in root. uncalcified, interglobular areas appear dark.



Ground section of outer dentine showing areas of interglobular dentine

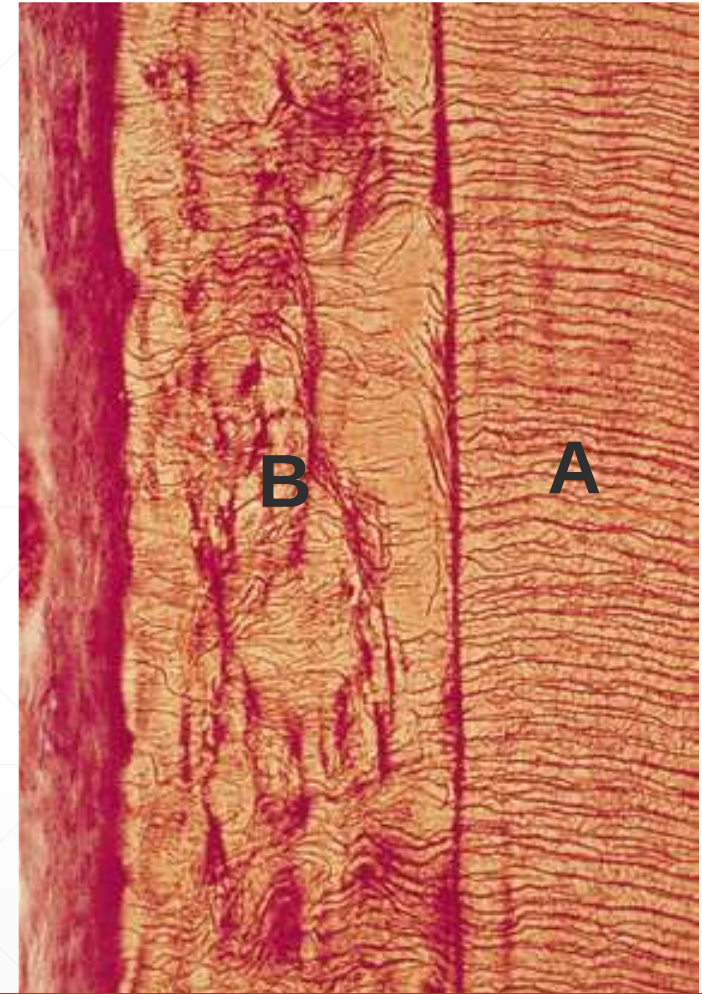
TERTIARY DENTINE

- Tertiary dentine is dentine formed in response to injury or stimuli: caries, cavity preparation, attrition, irritation, trauma.
- Newly-differentiated cells, which are responsible for tertiary dentine formation, are very similar to odontoblasts in that they produce type I collagen and dentine sialoprotein.



TERTIARY DENTINE

- **Reactionary Dentine:** mild stimulus, formed by surviving (pre-existing) odontoblasts. **Reparative Dentine:** severe stimulus, dentine is formed by newly differentiated odontoblast-like cells.
- It may resemble secondary dentine in having a regular tubular structure, or irregularly arranged tubules, or it may be relatively atubular.



Demineralised section showing loss of continuity of dentinal tubules between normal dentine (A) and tertiary dentine (B)

SCLEROTIC DENTINE

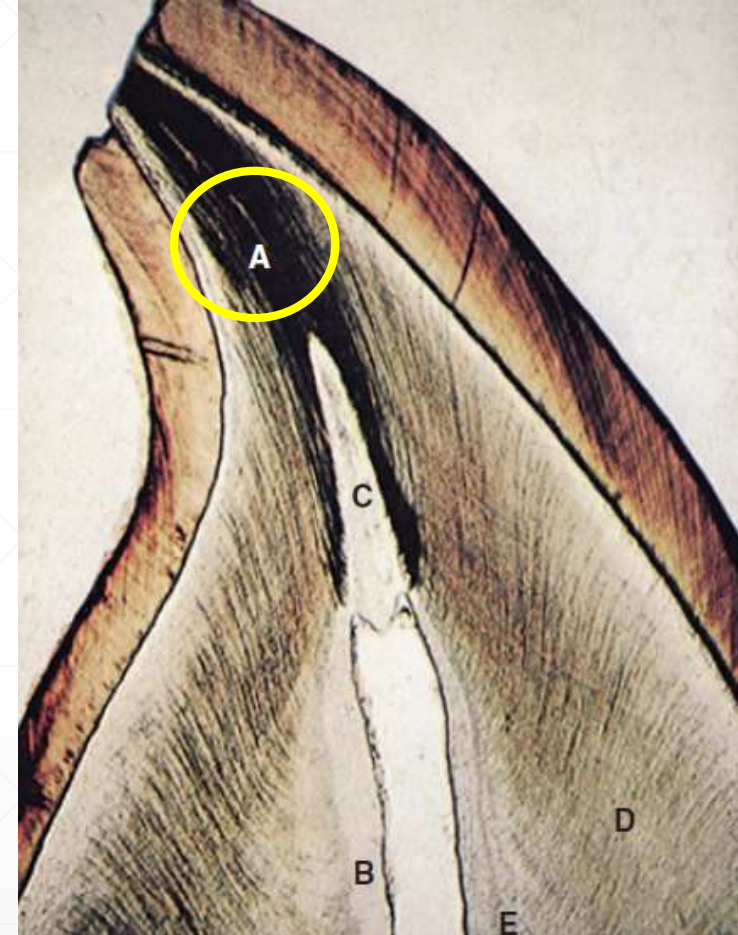
- The infilling with peritubular dentine as a physiological response to ageing eventually forms translucent dentine. When dentinal tubules fill in as a response to an external stimulus (caries or areas of attrition) is known as **sclerotic dentine**- also appearing translucent.



Longitudinal ground section showing loss of tubular structure and appearing transparent

DEAD TRACTS

- The areas in dentine where the odontoblastic processes have degenerated or died because of insult/trauma, or retract before peritubular dentine occludes the tubules, leaving the dentinal tubules empty.
- These empty tubules contain air in the ground sections and therefore appear dark.



Ground section showing dead tract (A)



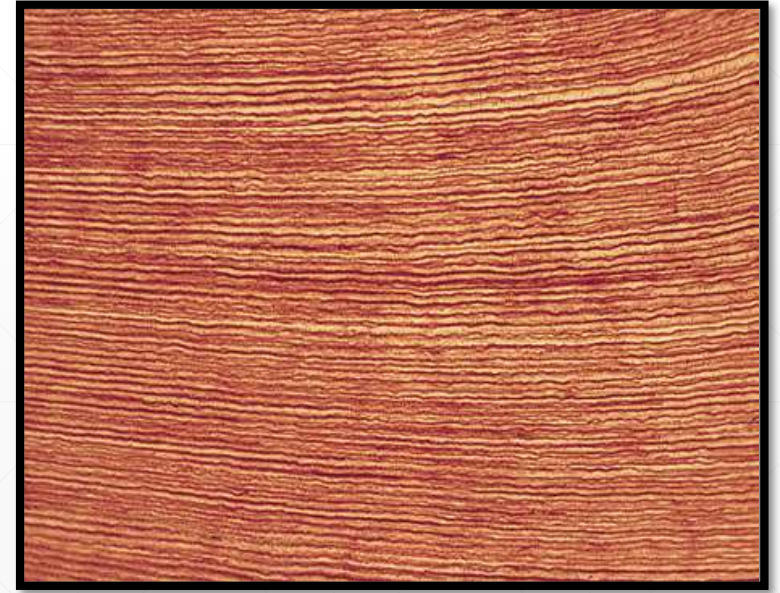
Dead tracts

Light micrograph showing dead tracts on the radicular carious lesion which appear dark under transmitted light

Primary curvatures: Due to the movement of odontoblasts after the formation of dentine, the tubules do not run straight. They follow S-shaped curvature



Secondary curvatures: Small, wave-like bends resulting from the lateral movement of odontoblasts. These are superimposed on the primary curvature.



STRUCTURAL LINES IN DENTINE

Structural lines in dentine are incremental or developmental lines formed during dentine deposition and mineralization.

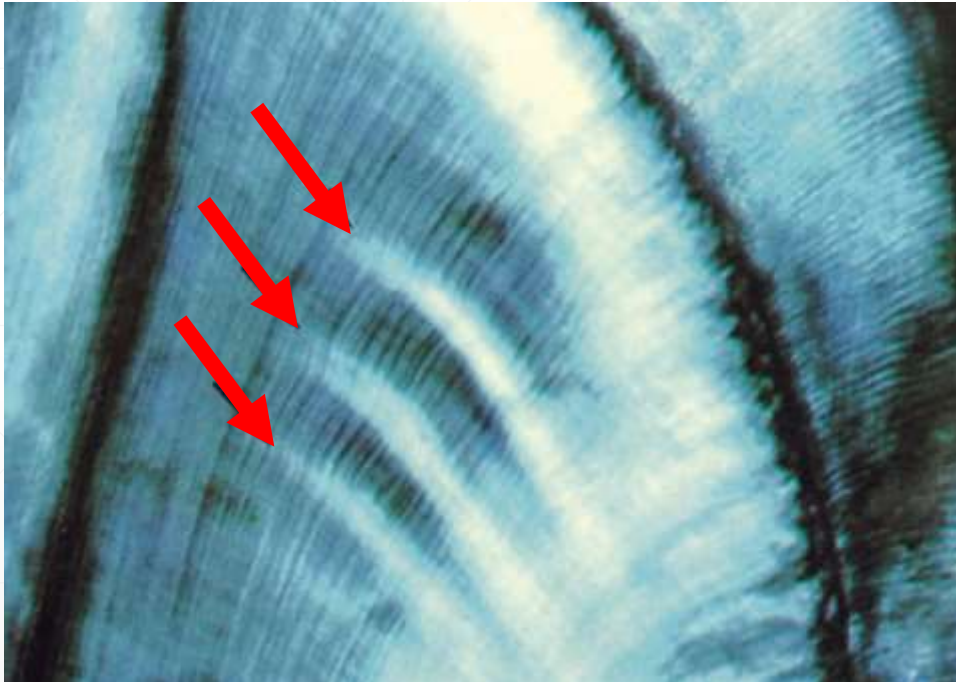
- Lines associated with the primary curvatures of the dentinal tubules
 - Lines associated with the secondary curvatures of the dentinal tubules
 - Incremental lines associated with matrix deposition and mineralisation
-

Lines associated with the primary curvatures of the dentinal tubules



When the sigmoid curvatures coincide, forming broad concentric bands in the circumpulpal dentine. These lines are known as **Schreger lines** as seen in (A) of a ground longitudinal section.

Lines associated with the secondary curvatures of the dentinal tubules



Contour lines of Owen are exaggerated lines found at the border of primary and secondary dentine due to coincidence of the secondary curvatures between neighboring dentinal tubules. They are caused by disturbances in mineralization.

Incremental lines associated with matrix deposition and mineralization

Dentine has regular, incremental, short-period, and long-period lines just like enamel.

Von Ebner's lines: Short-period lines, seen as alternating dark and light bands, each pair reflecting represent daily dentine deposition, 4 μm apart.

Andresen lines: The course long-period lines, representing slower rhythmic dentine deposition over longer intervals, 16–20 μm apart.

Between each long-period line, there are six to 10 pairs of short-period lines.

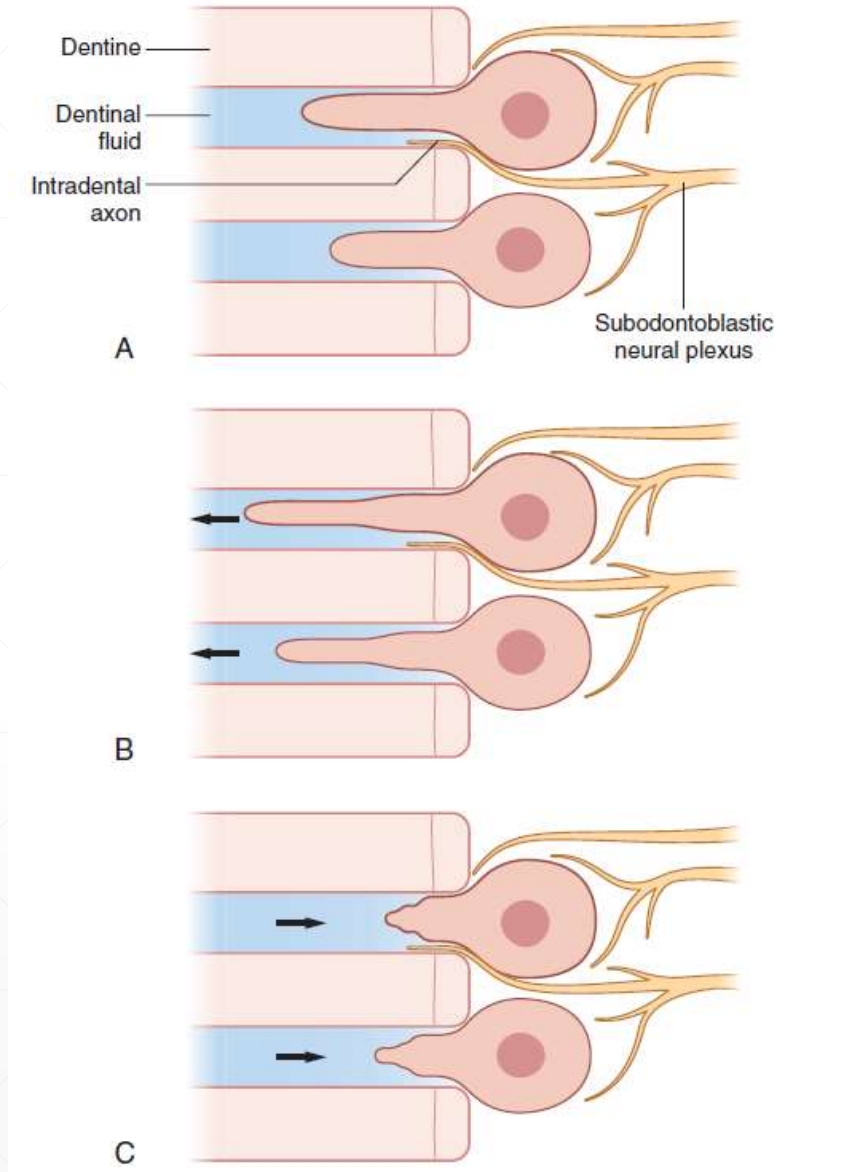
DENTINE SENSITIVITY

Exposed dentine is often sensitive. Three main hypotheses have been put forward to account for its sensitivity:

- nerves in dentine
 - the odontoblast processes
 - fluid movements in the dentinal tubules (hydrodynamic theory)
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DENTINE SENSITIVITY

Hydrodynamic Theory: The most accepted theory states that stimuli cause movement of fluid inside the dentinal tubules. This movement of fluid can depolarise nerve endings in the inner parts of tubules, at the pulp–predentine junction and subodontoblastic neural plexus, triggering sensation. Stimulus of cold, drying, and osmotic changes usually move fluid outward, while heat moves it inward.

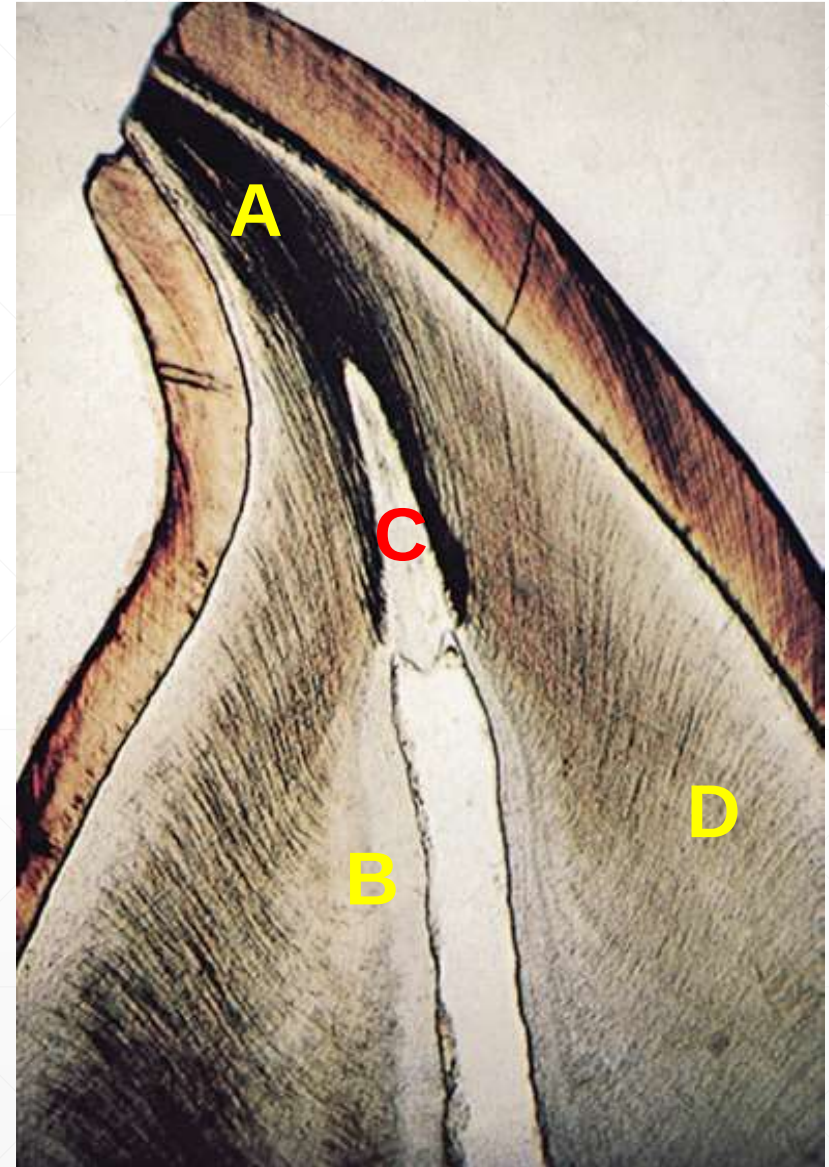


A- Dead tract

B- Secondary
Dentine

C- Tertiary Dentine

D- Circumpulpal
Dentine



THANK YOU!

